

Math Foundations

Machine learning is applied mathematics. This module covers the three areas you will meet in every later topic: linear algebra, derivatives and probability.

Linear Algebra Basics

Prerequisites: NumPy Basics.

A vector is an ordered list of numbers, such as a student's scores in five quizzes. A matrix is a grid of numbers with rows and columns; a dataset with 100 students and 5 features is a 100 by 5 matrix.

The dot product of two vectors multiplies matching elements and adds the results. It measures how much two vectors point in the same direction, and it is the core operation of linear models: a prediction is the dot product of a weight vector and a feature vector. Cosine similarity, used to compare text embeddings, is the dot product of two vectors divided by the product of their lengths.

Matrix multiplication combines rows of the first matrix with columns of the second, so an m by n matrix can only multiply an n by p matrix, giving an m by p result. The transpose flips a matrix over its diagonal. The identity matrix leaves any vector unchanged when multiplied.

The length of a vector is its norm. The L2 norm is the square root of the sum of squared elements. Norms appear again in regularization.

Derivatives

Prerequisites: none beyond school algebra.

A derivative measures how fast a function's output changes when its input changes a tiny amount. Geometrically it is the slope of the tangent line at a point. For $f(x) = x^2$, the derivative is $2x$, so at $x = 3$ the slope is 6.

Useful rules: the derivative of a constant is 0, the derivative of x to the power n is n times x to the power $n-1$, and the derivative of a sum is the sum of the derivatives.

When a function has several inputs, a partial derivative measures the change with respect to one input while holding the others fixed. Collecting all partial derivatives into a vector gives the gradient. The gradient points in the direction of steepest increase of the function.

Machine learning uses derivatives to answer one question: if I change this weight slightly, does the error go up or down, and by how much? That question is answered by gradient descent, and in neural networks by the chain rule and backpropagation.

Probability Basics

Prerequisites: none beyond school algebra.

Probability measures how likely an event is, on a scale from 0 (impossible) to 1 (certain). If an event A has probability 0.3, its complement "not A" has probability 0.7.

Two events are independent if one does not change the probability of the other; then $P(A \text{ and } B) = P(A) \text{ times } P(B)$. Conditional probability $P(A \text{ given } B)$ is the probability of A when we already know B happened. Bayes' theorem reverses conditionals: $P(A \text{ given } B) = P(B \text{ given } A) \text{ times } P(A) \text{ divided by } P(B)$. It is the basis of spam filters and many diagnostic tools.

A random variable takes numeric values with certain probabilities. Its expected value is the probability-weighted average. Variance measures spread around the expected value, and the standard deviation is the square root of the variance.

Classification models output probabilities, for example "0.92 probability this email is spam". Logistic regression and the cross-entropy loss used to train neural networks both rest on these ideas.